Data Dictionary — Olist Brazilian E-Commerce Dataset

Source: Kaggle — Brazilian E-Commerce Public Dataset by Olist
Period: October 2016 – October 2018
Total Orders: 99,441 | **Unique Customers:** 96,096 | **Files:** 9 CSVs

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Data Model Overview

```
olist_customers_dataset
        │
        └──▶ olist_orders_dataset (FACT TABLE)
                    │
        ┌───────────┼────────────────┐
        │           │                │
order_items    order_payments   order_reviews
        │
     products ──▶ category_translation
        │
     sellers ──▶ geolocation
```

File Descriptions

1. olist_orders_dataset.csv

Shape: 99,441 rows × 8 columns | **Role:** Central FACT table

Column	Type	Description	Nulls
`order_id`	string	Unique order identifier	0
`customer_id`	string	Key to customer dataset (not unique across orders)	0
`order_status`	string	delivered / shipped / canceled / unavailable / invoiced / processing / created / approved	0
`order_purchase_timestamp`	datetime	When customer placed the order	0
`order_approved_at`	datetime	When payment was approved	160 → filled with purchase timestamp
`order_delivered_carrier_date`	datetime	When order was handed to logistics partner	1,783 → undelivered orders
`order_delivered_customer_date`	datetime	When order was delivered to customer	2,965 → undelivered orders
`order_estimated_delivery_date`	datetime	Estimated delivery date shown to customer at purchase	0

Engineered columns:

- `delivery_delay_days` = `order_delivered_customer_date` − `order_estimated_delivery_date` (negative = early)
- `is_delivered` = 1 if delivered, 0 if not
- `Delay Bucket` = categorical grouping of delay (Early 7d+, Early 1-7d, On time, 1-3d late, etc.)

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2. olist_customers_dataset.csv

Shape: 99,441 rows × 5 columns

Column	Type	Description	Nulls
`customer_id`	string	Joins to orders dataset — NOT unique across orders	0
`customer_unique_id`	string	True unique customer identifier — use this for RFM	0
`customer_zip_code_prefix`	string	First 5 digits of zip code	0
`customer_city`	string	Customer city name	0
`customer_state`	string	2-letter Brazilian state code (e.g. SP, RJ)	0

> ⚠️ **Key note:** Always use `customer_unique_id` for customer-level analysis. `customer_id` repeats across orders and will overcount unique customers.

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3. olist_order_items_dataset.csv

Shape: 112,650 rows × 7 columns (multiple items per order)

Column	Type	Description	Nulls
`order_id`	string	Links to orders dataset	0
`order_item_id`	integer	Sequential item number within an order (1, 2, 3...)	0
`product_id`	string	Links to products dataset	0
`seller_id`	string	Links to sellers dataset	0
`shipping_limit_date`	datetime	Seller must ship by this date	0
`price`	float	Item price in BRL (R$)	0
`freight_value`	float	Freight cost charged to customer	0

Aggregated to order level:

- `total_items` = count of `order_item_id` per order
- `total_price` = sum of `price` per order

- `total_freight` = sum of `freight_value` per order

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4. olist_order_payments_dataset.csv

Shape: 103,886 rows × 5 columns (multiple payment methods per order)

Column	Type	Description	Nulls
`order_id`	string	Links to orders dataset	0
`payment_sequential`	integer	Multiple payments possible (1, 2, 3...)	0
`payment_type`	string	credit_card / boleto / voucher / debit_card	0
`payment_installments`	integer	Number of installments chosen	0
`payment_value`	float	Value paid in this transaction (BRL)	0

Aggregated to order level:

- `total_payment` = sum of `payment_value` per order
- `payment_installments` = max installments per order
- `payment_type` = first payment method per order

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5. olist_order_reviews_dataset.csv

Shape: 99,224 rows × 7 columns

Column	Type	Description	Nulls
`review_id`	string	Unique review identifier	0
`order_id`	string	Links to orders dataset	0
`review_score`	integer	Customer rating 1-5 (5 = best)	0
`review_comment_title`	string	Optional short title	87,656 → optional field
`review_comment_message`	string	Optional full review text	58,247 → optional field
`review_creation_date`	datetime	When review was created	0
`review_answer_timestamp`	datetime	When seller responded	0

> Only `review_score` is used in this project. Comment fields are optional and have high null rates — not a data quality issue.

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6. olist_products_dataset.csv

Shape: 32,951 rows × 9 columns

Column	Type	Description	Nulls
`product_id`	string	Unique product identifier	0
`product_category_name`	string	Category name in Portuguese	610 → filled as 'unknown'
`product_name_lenght`	integer	Character count of product name	610
`product_description_lenght`	integer	Character count of description	610
`product_photos_qty`	integer	Number of product photos	610
`product_weight_g`	float	Weight in grams	2 → filled with median
`product_length_cm`	float	Length in cm	2 → filled with median
`product_height_cm`	float	Height in cm	2 → filled with median
`product_width_cm`	float	Width in cm	2 → filled with median

7. olist_sellers_dataset.csv

Shape: 3,095 rows × 4 columns | **Nulls:** 0

Column	Type	Description
`seller_id`	string	Unique seller identifier
`seller_zip_code_prefix`	string	First 5 digits of seller zip
`seller_city`	string	Seller city
`seller_state`	string	2-letter state code

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8. olist_geolocation_dataset.csv

Shape: 1,000,163 rows × 5 columns | **Nulls:** 0

Column	Type	Description
`geolocation_zip_code_prefix`	string	Zip code prefix
`geolocation_lat`	float	Latitude
`geolocation_lng`	float	Longitude
`geolocation_city`	string	City name
`geolocation_state`	string	State code

> Used for geographic churn analysis. Contains multiple lat/lng per zip — use median or first record per zip.

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9. product_category_name_translation.csv

Shape: 71 rows × 2 columns | **Nulls:** 0

Column	Type	Description
`product_category_name`	string	Category name in Portuguese
`product_category_name_english`	string	English translation

Engineered / Master Table Columns

These columns were created during data cleaning and do not exist in the raw files:

Column	Source	Description
`delivery_delay_days`	orders	Actual − Estimated delivery date in days
`is_delivered`	orders	1 = delivered, 0 = not delivered
`Delay Bucket`	orders	Categorical delay grouping (7 values)
`Delay Sort`	orders	Numeric sort key for Delay Bucket (1–7)
`total_payment`	payments	Sum of all payments per order
`total_items`	items	Count of items per order
`total_price`	items	Sum of item prices per order
`total_freight`	items	Sum of freight per order
`product_category_name_english`	products + translation	English category name

RFM Table Columns

Column	Description
`customer_unique_id`	True unique customer ID

Column	Description
`Recency`	Days since last order from snapshot date
`Frequency`	Total number of orders placed
`Monetary`	Total spend in R$
`R_Score`	Recency score 1–5 (5 = most recent)
`F_Score`	Frequency score 1–5 (5 = most frequent)
`M_Score`	Monetary score 1–5 (5 = highest spend)
`Segment`	Champion / Loyal / Potential Loyalist / At Risk / Lost

Last updated: May 2026 | Project: Customer Churn & Revenue Leakage Analysis